



Turbidity and algal blooms

A report on behalf of the Wye Salmon Association.

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A brief note on turbidity measurement and the influence of illumination wavelength.

The data

At the beginning of July, there was a visible algal bloom on the River Wye, logged as such in EpiCollect. This was captured by the chlorophyll sensor on the EA sondes, which showed that the bloom was most intense upstream of the Lugg confluence and had pretty much fizzled out by Ross on Wye. But what caught my eye was the response on the turbidity signal from the sondes. It was barely perceptible.

Figure 1 shows the chlorophyll signal¹ from Hampton Bishop, located downstream of Hereford but above the Lugg confluence. It shows two humps reaching circa 130 fu. For reference, EA would consider a genuine bloom to be anything over about 30 fu.

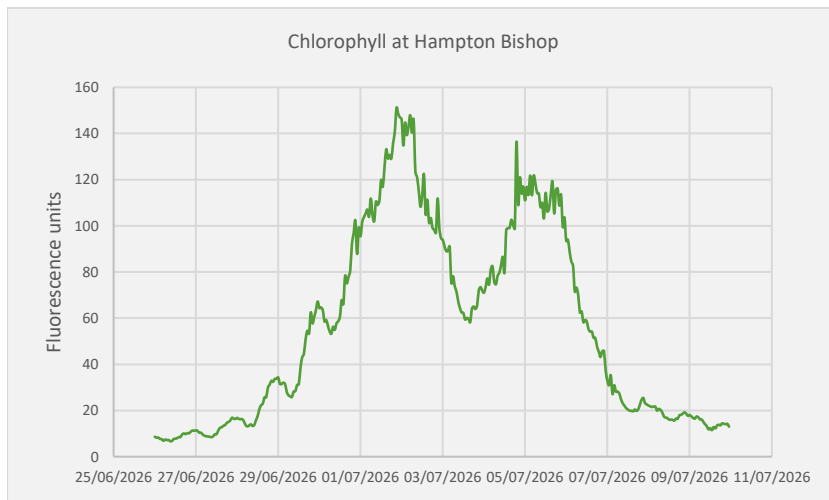


Figure 1: chlorophyll signal from the EA sonde at Hampton Bishop

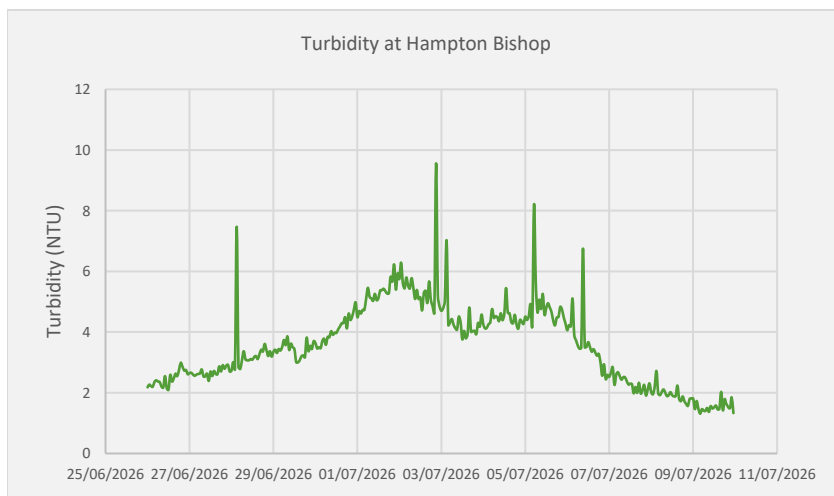


Figure 2: turbidity signal from the EA sonde at Hampton Bishop.

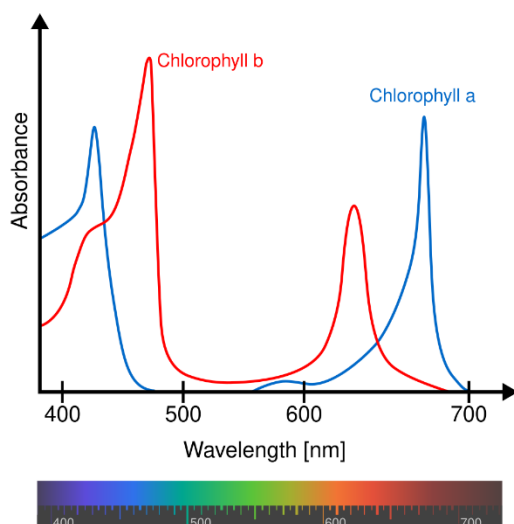
Now, looking at turbidity over the same period (figure 2), we can clearly see two corresponding peaks, but they are very weak, barely rising 4 NTU above background. Compare this to the turbidity signals seen on the 2025 CS data from the Worm sub catchment (Worm Brook 2025). We recorded visible blooms at several locations that summer, with turbidity rising as high as 95 NTU during a visible bloom. Figure 3 shows a typical plot from the Jury Brook site, where turbidity remained above 50 NTU throughout June, peaking at 82 NTU. Whilst the EA and CS datasets are obviously from different times and places, the CS turbidity data, measured on a WSA turbidity meter, shows an order of magnitude larger response than does the EA data. So, why the difference?

¹ Chlorophyll is detected by means of its fluorescence. A blue LED is used to excite the fluorescence, and a detector measures the resulting red emission. Calibration against actual algal density is fraught with difficulties and calibration is more usually performed using a standard solution of rhodamine B (a synthetic dye with similar absorption/emission characteristics). Unit of measure is thus an arbitrary "fluorescence unit".

Measuring turbidity

Both standards require calibration against the same standard formazin suspension, so necessarily agree for this material. For practical situations, such as suspended mineral loads in swollen rivers, they inevitably diverge. Whilst differences are unlikely to be hugely significant, we always need to keep them in mind. But with optically active materials such as chlorophyll, things get more interesting.

Chlorophyll



But chlorophyll does not absorb IR light (above 700nm). Therefore, chlorophyll is substantially transparent under IR illumination. Green algal cells are thus largely invisible to a turbidity sensor which uses IR light (830-890nm being mandated by ISO 7027-1).

Conclusion

The spectral absorption characteristics of chlorophyll result in a substantial difference in sensitivity to green algae between the WSA turbidity meter (visible light) and the EA sonde's turbidity sensor (IR light). The absence of absorption in the near IR cause chlorophyll containing algae to be substantially invisible to the EA sonde. For the EA, this lack of chlorophyll sensitivity on the turbidity signal is overcome by also having a specific chlorophyll fluorescence sensor.

Conversely, the chlorophyll absorption bands in the visible spectrum mean that the WSA turbidity meter effectively detects chlorophyll containing algae, broadly correlating with what we see with our eyes, or might measure with a Secchi tube. Chlorophyll fluorescence sensor not required.